

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – IV

Roll No.: T006	Name: Nandini Chaudhary
Class: TYIT	Batch:01
Date of Assignment: 20-08-2026	Date/Time of Submission:20-08-2026

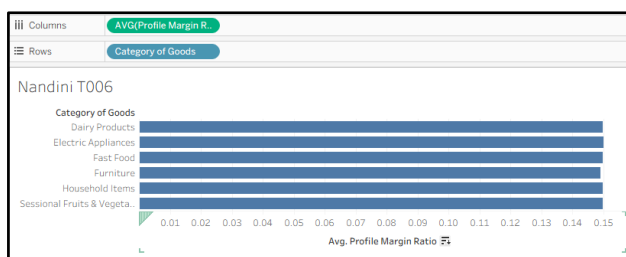
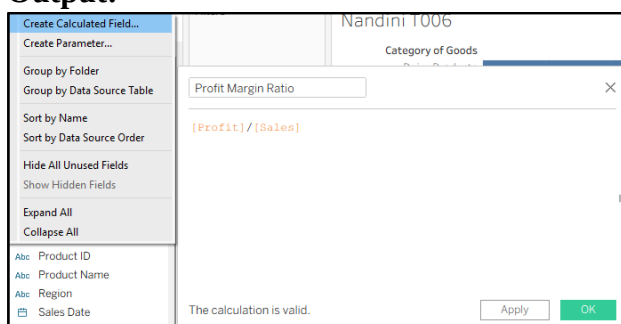
Performing Data Analysis using Calculations in Tableau

A) Create basic calculated fields using formulas.

Steps:-

- 1.Open the dataset in Tableau.
- 2.In the Data pane, right-click on an empty area.
- 3.Select Create Calculated Field.
- 4.Enter the name Profit Margin Ratio.
- 5.Write the formula:[Profit] / [Sales]
- 6.Check that Tableau shows “The calculation is valid.” Click OK.
- 7.Drag Profit Margin Ratio to Columns.
- 8.Drag Category of Goods to Rows.
- 9.The Bar Chart will be displayed.

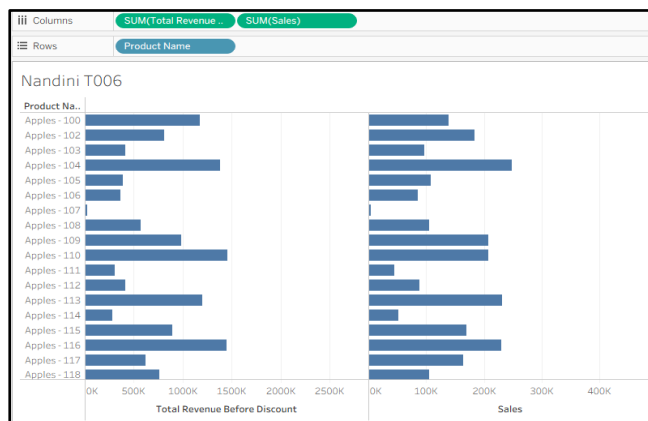
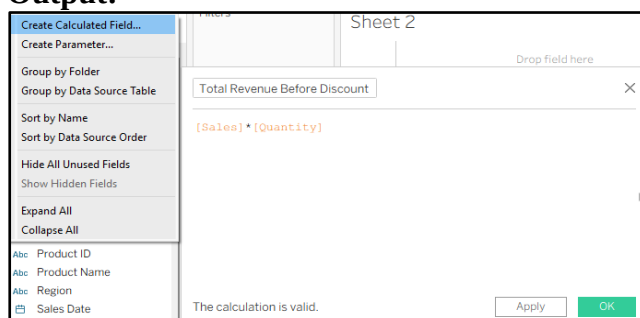
Output:-



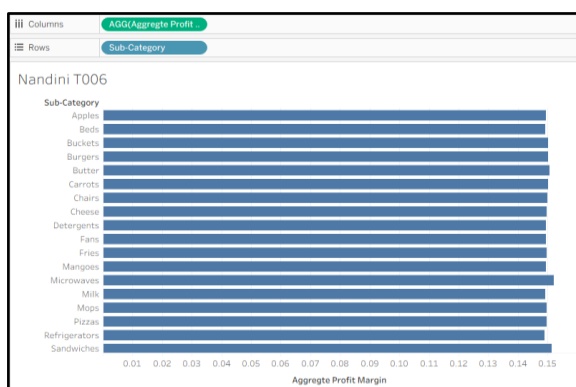
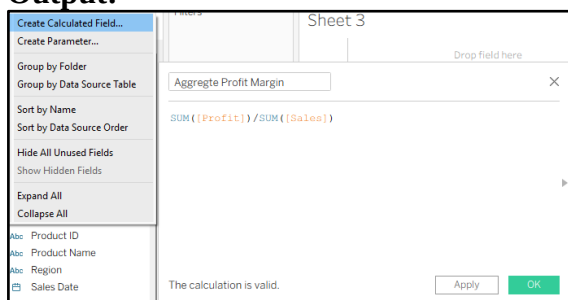
B)Perform row-level calculations.

Steps:-

- 1.Right-click in the Data pane.
- 2.Select Create Calculated Field.
- 3.Enter the name Total Revenue Before Discount.
- 4.Write the formula: [Sales] * [Quantity]
- 5.Check that the calculation is valid. Click OK.
- 6.Drag Product Name to Rows.
- 7.Drag Total Revenue Before Discount to Columns.
- 8.Drag Sales also to Columns.
- 9.Tableau will display the comparison of Total Revenue Before Discount and Sales.

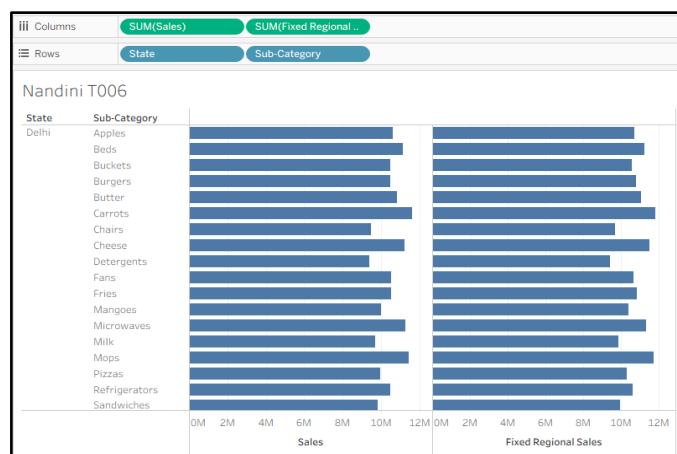
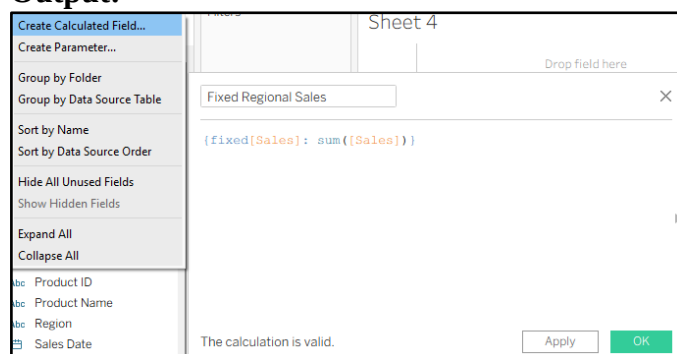
Output:-**C) Perform aggregate-level calculations.****Steps:-**

- 1.Right-click in the Data pane.
- 2.Select Create Calculated Field.
- 3.Enter the name Aggregate Profit Margin.
- 4.Write the formula: **SUM([Profit]) / SUM([Sales])**
- 5.Check that “The calculation is valid.” is displayed. Click OK.
- 6.Drag Aggregate Profit Margin to Columns.
- 7.Drag Sub-Category to Rows.
- 8.The aggregate profit margin bar chart will be displayed.

Output:-

d) Implement FIXED Level of Detail (LOD) calculations.**Steps:-**

- 1.Right-click in the Data pane.
- 2.Select Create Calculated Field.
- 3.Enter the name Fixed Regional Sales.
- 4.Write the formula: **{ FIXED [Sales] : SUM([Sales]) }**
- 5.Check that the calculation is valid. Click OK.
- 6.Drag Sales to Columns.
- 7.Drag Fixed Regional Sales to Columns.
- 8.Drag State to Rows.
- 9.Drag Sub-Category to Rows.
- 10.The comparison between Sales and Fixed Regional Sales will be displayed.

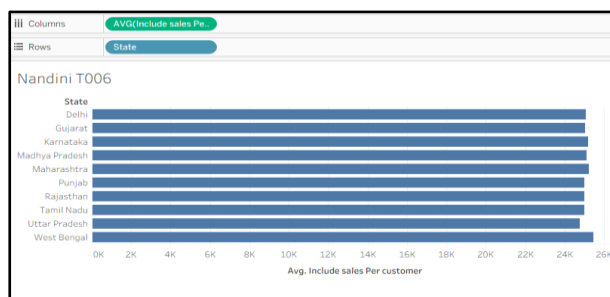
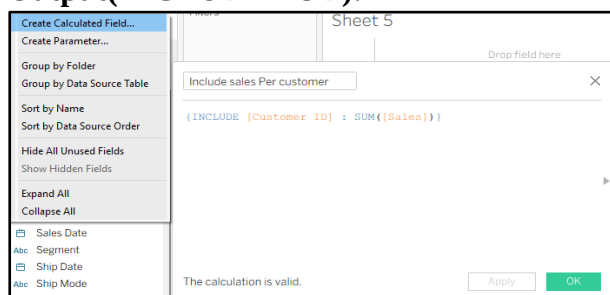
Output:-**E) Use INCLUDE and EXCLUDE LOD expressions.****Steps:-****INCLUDE LOD**

- 1.Right-click the Data pane → Create Calculated Field.
- 2.Name it Include sales per customer.
- 3Formula: **{ INCLUDE [Customer ID] : SUM([Sales]) }**
- 4.Click OK and drag it to Columns.
- 5.Drag State to Rows.

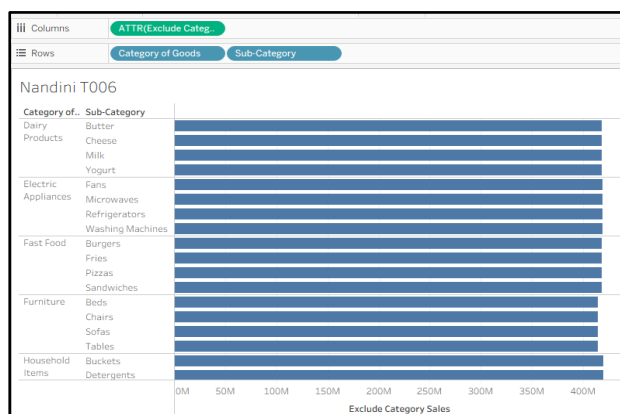
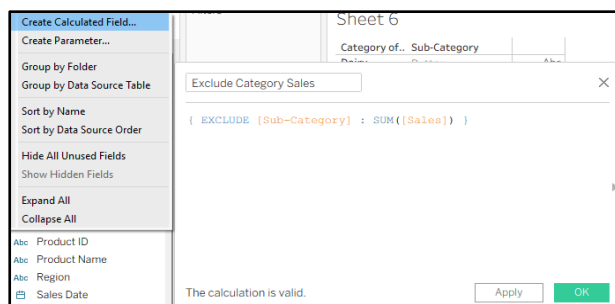
EXCLUDE LOD

- 1.Right-click the Data pane → Create Calculated Field.
- 2.Name it Exclude Category Sales.
- 3.Formula: **{ EXCLUDE [Sub-Category] : SUM([Sales]) }**
- 4.Click OK and drag it to Columns.
- 5.Drag Category of Goods and Sub-Category to Rows.

Output(INCLUDE LOD):-



EXCLUDE LOD:-



f) Create and use parameters for dynamic analysis

Steps:-

1. In the Data pane, right-click on an empty area.
2. Select Create Parameter.
3. Enter the name select Profit Threshold.
4. Set Data Type as Float.
5. Set the Current Value to 500.
6. Keep Allowable Values as All.
7. Keep Value when the workbook opens as Current value. Click OK.
8. The parameter can now be used in a calculated field for dynamic analysis.

Parameter Details:

Name: select Profit Threshold

Data Type: Float

Current Value: 500

Allowable Values: All

Output:-

Create Parameter

Name
select Profit Threshold

Properties
Data type: Float, Display format: 500

Current value: 500, Value when workbook opens: Current value

Allowable values: ☒ All, ☐ List, ☐ Range

Buttons: Cancel, OK

Create Calculated Field...

Name: KPI Threshold Met?

Formula: If SUM([Profit]) >= [select Profit Threshold] THEN "Target Achieved" ELSE "Below Target" END

The calculation is valid.

Buttons: Apply, OK



g) Apply calculations to fix data issues and enhance insights.

Steps:-

- 1.Right-click the Data pane → Create Calculated Field.
- 2.Name it KPI Threshold Met?
- 3.Enter the formula: **IF SUM([Profit]) >= [select Profit Threshold] THEN "Target Achieved" ELSE "Below Target" END**
- 4.Click OK.
- 5.Drag Profit to the view and State to Rows.
- 6.Use KPI Threshold Met? to show Target Achieved or Below Target.

Output:-

Create Calculated Field...

Name: cleaned Discount

Formula: IFNULL([Discount], 0)

The calculation is valid. 1 Dependency

Buttons: Apply, OK

